AI/ML PROJECT SUBMISSION

PROJECT OVERVIEW	STUDENT DETAILS
• Project Number: 1 • Task Name: Student Management System. • Difficulty: Easy	• Name: Pranav Gajanan Rane • College: Government Polytechnic Gondia • Branch: Computer Engineering • Domain: AI/ML

```
def add_student() : # Adds Students data
 try :
 roll_no = int(input("Enter the roll_no – "))
 if roll_no in student_records :
 print("Rollno already exists!")
 else :

 name = input("Enter your name – ")
 if name.isdigit():
 print("Name can only be in string!")
 else :
 mark1 = int(input("Enter your marks in Science – "))
 mark2 = int(input("Enter your marks in Social Science
– "))
 mark3 = int(input("Enter your marks in Maths – "))
 mark4 = int(input("Enter your marks in Hindi/Sanskrit
– "))
 mark5 = int(input("Enter your marks in English – "))


          if (mark1 > 100 or mark2 > 100 or mark3 > 100 or
mark4 > 100 or mark5 > 100) or \
         (mark1 < 0 or mark2 < 0 or mark3 < 0 or mark4 < 0
or mark5 < 0):
          print("Marks must be between 0 and 100!")
        else:
          total = (mark1 + mark2 + mark3 + mark4 + mark5)

          percentage, grade = calculate_grade(total)
```

```
 student_records[roll_no] = {
 "name" : name,
 "Science_Marks" : mark1,
 "Social_Science_Marks" : mark2,
 "Maths_Marks" : mark3,
 "H/S_Marks" : mark4,
 "English_Marks" : mark5,
 "percentage" : percentage,
 "grade" : grade
 }
except ValueError :
   print("Enter correct Value!")

def view_all() : # Used to view all the students details
if any.
  if not student_records :
    print("Data doesn't exist")
  else :
    fmt = "{:<8} {:<15} {:<8} {:<8} {:<8} {:<8} {:<8} {:<12} {:
<6}"
    print("=" * 105)
      print(fmt.format("Roll No", "Name", "Sci", "SocSci",
"Math", "Hin/San", "Eng", "Percentage", "Grade"))
    print("=" * 105)

    for roll_no,details in student_records.items() :
      print(fmt.format(
        roll_no,
        details["name"],
        details["Science_Marks"],
        details["Social_Science_Marks"],
```

```
 details["Maths_Marks"],
 details["H/S_Marks"],
 details["English_Marks"],
  f"{details['percentage']:.2f}%",  #  rounds  percentage
upto 2 decimal places
 details["grade"]
 ))
 print("=" * 105)


def  search_student()  :  #  Searchs  student  a  student
by their roll_no
 try :
 if not student_records :
 print("Data doesn't exist")
 else :
 roll_no = int(input("Enter a rollno to view its data - "))
 if roll_no in student_records :
 details = student_records[roll_no]
 print("\n" + "=" * 40)
 print(f"   FOUND RECORD (Roll No: {roll_no})")
        print("=" * 40)
        print(f" Name:    {details['name']}")
        print(f" Science:  {details['Science_Marks']}")
                                                print(f"      Social      Sci:
{details['Social_Science_Marks']}")
        print(f" Maths:   {details['Maths_Marks']}")
        print(f" Hindi/Sans: {details['H/S_Marks']}")
        print(f" English:  {details['English_Marks']}")
        print("-" * 40)
        print(f" Percentage: {details['percentage']:.2f}%")
```

```
print(f" Grade: {details['grade']}")
print("=" * 40)
else :
print("Record not found for this key!")
except :
print("Pls enter correct values!")


def delete_student() : # Removes a students data
  try :
    roll_no = int(input("Enter the roll no whose data you
want to remove - "))
    if roll_no in student_records :
      print(f"Removing roll_no {roll_no}'s data.")
      del student_records[roll_no]
      print("Data Removed!")
    else :
      print("Roll no does not exists")
  except ValueError :
    print("Pls enter correct value!")


def calculate_grade(total) : # calculates grade and
send back to add add_student()
  percentage = (total / 500) * 100
  grade = ""
  if percentage >= 90 :
    grade = "A"
  elif percentage >= 80 :
    grade = "B"
  elif percentage >= 70 :
```

```
 grade = "C"
 elif percentage >= 60 :
 grade = "D"
 elif percentage >= 50 :
 grade = "E"
 else :
 grade = "F"
 return percentage,grade
def update_student() : #updates any required values
of student
  try :
    roll_no = int(input("Enter a rollno to change its data
- "))
   if roll_no in student_records :

     print("What do you want to change - ")
     print("1)Name")
     print("2)Marks of Science")
     print("3)Marks of Math")
     print("4)Marks of Social Science")
     print("5)Marks of Hindi/Sanskrit")
     print("6)Marks of English ")

     choice = input("Enter your choice (1 - 6) -")
     if choice == "1" :
       name = input("Change the name to - ")
       if name.isdigit():
         print("Name can only be in string!")
       else :
         student_records[roll_no]['name'] = name
         print("Name updated")
```

```
def update_student() : #updates any required values
of student
  try :
    roll_no = int(input("Enter a rollno to change its data
- "))
    if roll_no in student_records :

      print("What do you want to change - ")
      print("1)Name")
      print("2)Marks of Science")
      print("3)Marks of Math")
      print("4)Marks of Social Science")
      print("5)Marks of Hindi/Sanskrit")
      print("6)Marks of English ")

      choice = input("Enter your choice (1 - 6) -")
      if choice == "1" :
        name = input("Change the name to - ")
        if name.isdigit():
          print("Name can only be in string!")
        else :
          student_records[roll_no]['name'] = name
          print("Name updated")
      elif choice == "2" :
        mark1 = int(input("Enter updated marks - "))
              student_records[roll_no]['Science_Marks']  =
mark1
        print("Marks Updated!")
```

```
elif choice == "3" :
        mark1 = int(input("Enter updated marks - "))
        student_records[roll_no]['Maths_Marks'] = mark1
        print("Marks Updated!")
      elif choice == "4" :
        mark1 = int(input("Enter updated marks - "))
                                             student_records[roll_no]
['Social_Science_Marks'] = mark1
        print("Marks Updated!")
      elif choice == "5" :
        mark1 = int(input("Enter updated marks - "))
        student_records[roll_no]['H/S_Marks'] = mark1
        print("Marks Updated!")
      elif choice == "6" :
        mark1 = int(input("Enter updated marks - "))
                  student_records[roll_no]['English_Marks']  =
mark1
        print("Marks Updated!")
      else :
        print("Pls Enter the correct value(1-6)")
       total = student_records[roll_no]['Science_Marks']
+ student_records[roll_no]['Social_Science_Marks'] +
student_records[roll_no]['Maths_Marks']             +
student_records[roll_no]['English_Marks']           +
student_records[roll_no]['H/S_Marks']
      percentage, grade = calculate_grade(total)
                      student_records[roll_no]['percentage']   =
percentage
      student_records[roll_no]['grade'] = grade
```

```
else :
print("The rollno doesnt exist!")
except ValueError :
print("Pls enter correct values!")

def class_report(): # Bonus Task
 if not student_records:
   print("No data to generate report!")
   return
 else:
   #Calculating topper
   highest = -1.0
   name = ""

   for key,value in student_records.items() :
     pct = value['percentage']

     if pct > highest :
       highest = pct
       name = value['name']
   print("*"*10)
      print(f"{name}  achieved  highest  percentage  -
{highest}")
   print("*"*10)


   #Finding Average Percentage
   all_students = len(student_records)
   total_percentage = 0
   for key,value in student_records.items():
```

```
total_percentage += value['percentage']
avg_percentage = total_percentage / all_students
print(f"Average Percentage are - {avg_percentage}")
print("*"*10)

  #Pass/Fail
  passed = 0
  failed = 0
  for key,value in student_records.items():
    if value['percentage'] < 50 :
      failed += 1
    else :
      passed += 1
  print(f"Total student passed are - {passed}")
  print(f"Total Student Failed are - {failed}")



def show_menu() : #Shows what you can perform
  print("What you can perfrom - ")
  print("1)Add a Student")
  print("2)View All Students")
  print("3)Search a student by their Roll Number")
  print("4)Update a student record")
  print("5)Remove a student data")
  print("6)See Class Report")
  print("7)Exit")


'''Actual System Below'''
```

```
student_records = {} #(Global Declaration) Holds students details

def main() :
  while True :
    try :
      print("-" * 58)
        print("_____-Welcome to Student Management System!_____")
        print("Note - Asks for user input to perform a specific action.")
      print("-" * 58)

      show_menu()

       choice = input("Enter what you want to do (1 - 7) - ")
      print("-" * 30)
      if choice == "1" :
        add_student()
      elif choice == "2" :
        view_all()
      elif choice == "3" :
        search_student()
      elif choice == "4" :
        update_student()
      elif choice == "5" :
        delete_student()
      elif choice == "6" :
```

```
 class_report()
 elif choice == "7" :
 print("Exited Successfully!")
 break
 else :
print("Pls enter a valid choice(1 - 7)")
   except ValueError :
     print("Enter correct value!")

if __name__ == "__main__":
  main()
```

Note - This code might break due to incorrect indentation while pdf making so pls use the code mentioned in .py file attached

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TEST OUTPUTS -

```
----------------------------------------------------------
_____-Welcome to Student Management System!_____
Note - Asks for user input to perform a specific action.
----------------------------------------------------------
What you can perfrom -
1)Add a Student
2)View All Students
3)Search a student by their Roll Number
4)Update a student record
5)Remove a student data
6)See Class Report
7)Exit
Enter what you want to do (1 - 7) -
```

```
Enter what you want to do (1 - 7) - 1
------------------------------
Enter the roll_no - 101
Enter your name - Pranav
Enter your marks in Science - 98
Enter your marks in Social Science - 97
Enter your marks in Maths - 96
Enter your marks in Hindi/Sanskrit - 95
Enter your marks in English - 85
```

```
Enter what you want to do (1 - 7) - 2
------------------------------
==========================================================================================
Roll No  Name           Sci      SocSci   Math     Hin/San  Eng      Percentage   Grade
==========================================================================================
101      Pranav         98       97       96       95       85       94.20%       A
102      Gauri          95       75       85       95       75       85.00%       B
103      Ishant         23       42       32       16       23       27.20%       F
==========================================================================================
```

```
Enter what you want to do (1 - 7) - 3
------------------------------
Enter a rollno to view its data - 101

========================================
       FOUND RECORD (Roll No: 101)
========================================
 Name:         Pranav
 Science:      98
 Social Sci:   97
 Maths:        96
 Hindi/Sans:   95
 English:      85
----------------------------------------
 Percentage:   94.20%
 Grade:        A
========================================
```

```
Enter what you want to do (1 - 7) - 6
------------------------------
**********
Pranav achieved highest percentage - 94.19999999999999
**********
Average Percentage are - 68.8
**********
Total student passed are - 2
Total Student Failed are - 1
```

TEST OUTPUTS IN BROKEN CASES -

```
Enter what you want to do (1 - 7) - 1
------------------------------
Enter the roll_no - Pranav
Enter correct Value!
```

```
Enter what you want to do (1 - 7) - 3
------------------------------
Enter a rollno to view its data - 95
Record not found for this key!
```

```
Enter what you want to do (1 - 7) - 5
------------------------------
Enter the roll no whose data you want to remove - 95
Roll no does not exists
```

```
Enter what you want to do (1 - 7) - 5
------------------------------
Enter the roll no whose data you want to remove - Pranav
Pls enter correct value!
----------------------------------------------------------
```

CODE TESTING WAS DONE SUCCESSFULLY!

More Details about the project is mentioned in the .md file attached . This pdf was made specifically to attach the code and its outputs.

PROJECT COMPLETED